Mental Health Assessment and Analysis System

21CSC205P Database Management Systems

A MINI PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The Mental Health Assessment and Analysis System is a database-driven application designed to collect user responses to psychological assessment questions, calculate total scores, and determine mental health patterns based on predefined score ranges. The system uses ER modeling and relational database concepts to manage users, assessments, responses, and result patterns efficiently.

The project demonstrates the implementation of database design principles including ER modeling, relational schema design, normalization techniques, SQL queries, triggers, transactions, concurrency control, and recovery mechanisms. The system ensures data integrity, consistency, and efficient retrieval of assessment results.

This project highlights the practical application of DBMS concepts in building a structured and normalized database system for real-world assessment analysis.

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CHAPTER 1

PROBLEM UNDERSTANDING, IDENTIFICATION OF ENTITY AND RELATIONSHIPS, CONSTRUCTION OF DB USING ER MODEL FOR THE Mental Health Assessment and Analysis System

1.1 Introduction

Mental health assessment is an important process for evaluating an individual's psychological and emotional well-being. With increasing academic, professional, and social pressures, there is a growing need for structured systems that can collect and analyze assessment data accurately. Manual methods of recording responses are often inefficient, prone to errors in score calculation, and lack proper organization for long-term data management. A database-driven approach provides a systematic solution for storing, processing, and retrieving assessment information in a consistent and reliable manner.

The proposed *Mental Health Assessment and Analysis System* focuses on designing and implementing a relational database that records user responses to predefined mental health questions, calculates total scores, and classifies results based on predefined score ranges. The project emphasizes the application of core Database Management System (DBMS) concepts such as Entity-Relationship modeling, relational schema design, normalization, and SQL implementation. The system ensures data integrity, eliminates redundancy, and enables efficient retrieval of results, thereby demonstrating the practical use of structured database design in an assessment-based application.

1.2 Motivation

The motivation behind this project is to design a structured and efficient database system that eliminates the limitations of manual mental health assessment processes. Manual methods often lead to data redundancy, calculation errors, and difficulty in maintaining consistent records. By implementing a relational database system, assessment data can be stored systematically, ensuring accuracy in score computation and reliable classification of mental health patterns. Additionally, this project provides an opportunity to apply fundamental Database Management System concepts such as ER modeling, relational schema design, normalization, and SQL implementation in a practical and real-world context. The system aims to demonstrate how structured data management can enhance reliability, consistency, and efficiency in assessment-based applications.

1.3 Scope

The scope of the Mental Health Analyzer is limited to database design and implementation using DBMS principles.
The scope includes:

- User registration and management.
- Question storage.
- Recording responses.
- Automatic score calculation.
- Mapping total score to mental health category.
- Maintaining normalized database structure.

The project primarily focuses on backend database design, which serves as the foundation for future system enhancements.

1.4 Problem Statement

The primary problem addressed in this project is the lack of a structured and reliable system for storing, managing, and analyzing mental health assessment data. Manual or unorganized methods of recording responses often result in data redundancy, inconsistencies, and errors in score calculation. Furthermore, without a properly designed database, it becomes difficult to maintain relationships between users, assessments, responses, and result classifications. This creates challenges in ensuring data integrity and efficient retrieval of information.

The objective of this project is to design and implement a relational database system that systematically records user responses to predefined mental health questions, calculates total assessment scores, and maps those scores to predefined mental health patterns. The system must maintain referential integrity using primary and foreign keys, follow normalization principles to eliminate redundancy, and support efficient query processing. The project aims to demonstrate the practical application of DBMS concepts in building a structured and consistent assessment-based data management system.

1.5 Project Requirements

The proposed *Mental Health Assessment and Analysis System* requires the development of a structured relational database capable of storing, managing, and analyzing assessment data efficiently. The system must ensure proper organization of data entities and maintain integrity through well-defined relationships. The requirements of the project can be categorized as follows:

Functional Requirements

The system must support the following operations:

- Storage of user details with a unique identifier for each user.
- Maintenance of predefined mental health assessment questions.
- Recording of user responses for each assessment attempt.
- Calculation and storage of total assessment scores.
- Classification of assessment results based on predefined score ranges.

Non-Functional Requirements

In addition to functional capabilities, the system must satisfy the following quality requirements:

- Enforcement of primary key and foreign key constraints to maintain referential integrity.
- Prevention of invalid or duplicate data entries.
- Maintenance of consistency during score calculation and data storage.
- Efficient retrieval of assessment results through structured SQL queries.

The database design must follow normalization principles to eliminate redundancy and ensure logical data organization. The overall implementation should demonstrate the practical application of core DBMS concepts such as ER modeling, relational schema design, and structured query execution.

1.6 Identification of Entity and Relationships

Entities:

- USER
- ASSESSMENT
- QUESTION
- RESPONSE
- MENTAL_HEALTH_PATTERN

Relationships:

- USER takes ASSESSMENT
- ASSESSMENT contains RESPONSE

- QUESTION answered in RESPONSE
- ASSESSMENT generates MENTAL_HEALTH_PATTERN

Cardinality:

- One USER → many ASSESSMENTS (1:M)
- One ASSESSMENT → many RESPONSES (1:M)
- One QUESTION → many RESPONSES (1:M)

1.7 Construction of DB Using ER Model for the Mental Health Assessment and Analysis System

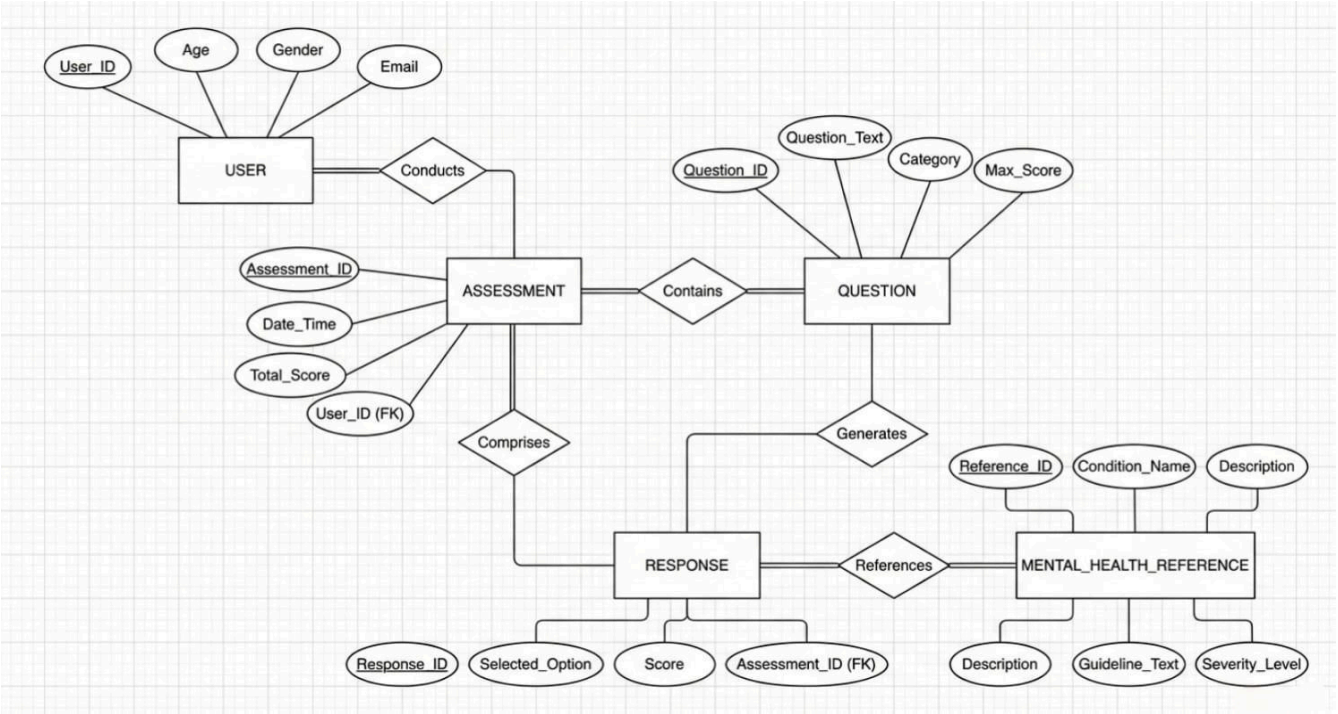

Figure 1.1: ER Diagram of Mental health Assessment and Analysis System

Chapter 2

Design of Relational Schemas, Creation of Database and Tables for Mental Health Assessment and Analysis System

2.1 Relational Schema for Mental Health Assessment and Analysis System

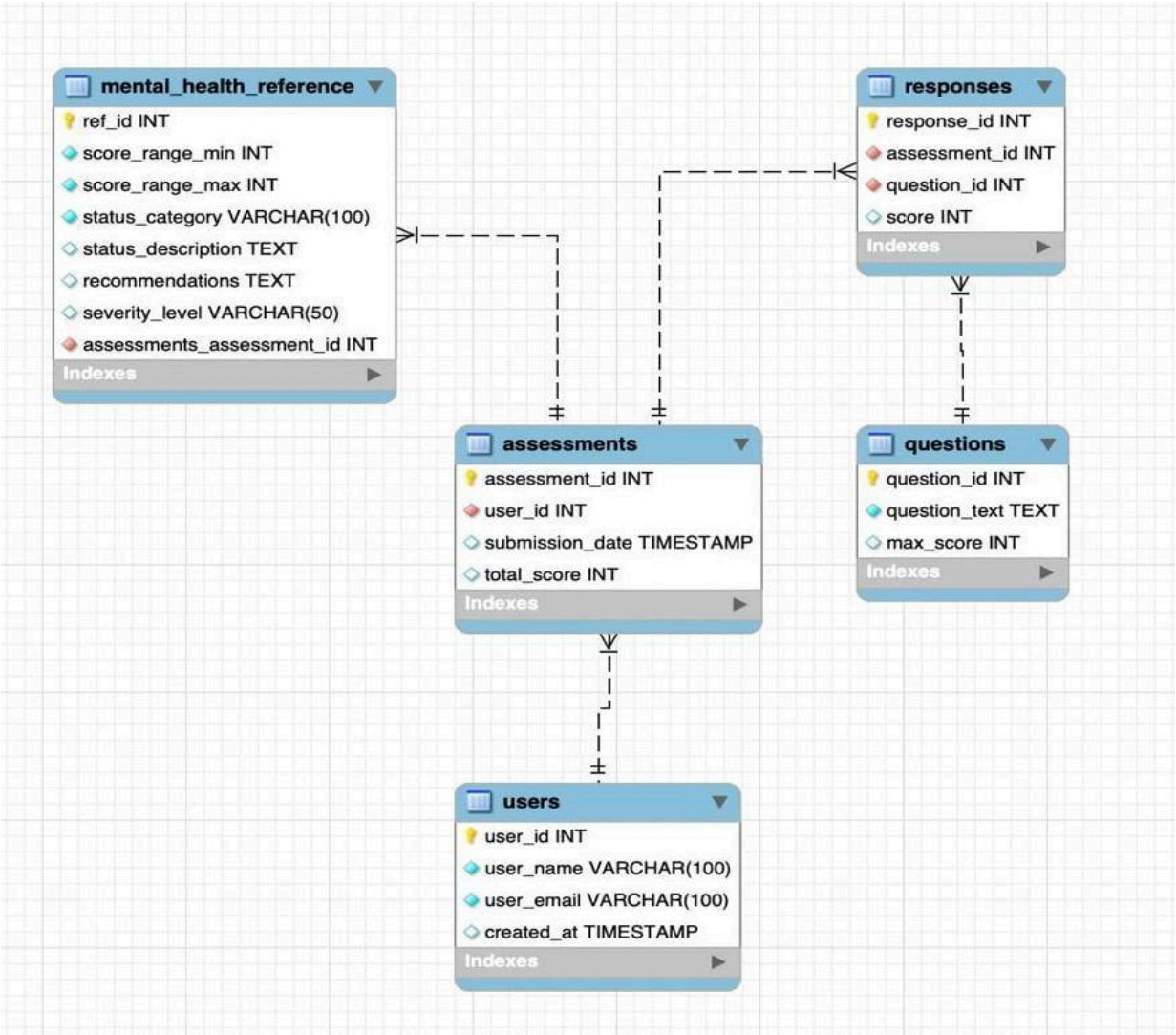

Figure 2.1: Relational Schema for Mental Health Assessment and Analysis System

2.2 Description of Tables

The tables used in the Mental Health Assessment and Analysis database along with their attributes, data types, size, and constraints are described below.

Table 2.1: User

- **Purpose:** Stores information about the user.
- **Primary Key:** id(Integer)
- **Data Types and Constraints:**

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
user_name	varchar(100)	NO		NULL	
user_email	varchar(100)	NO	UNI	NULL	
created_at	timestamp	YES		CURRENT	DEFAULT

Table 2.2: Responses

- **Purpose:** To collect the responses of the questions.
- **Primary Key:** response_id(Integer)
- **Data Types and Constraints:**

Field	type	Null	Key	Default	Extra
response_id	int	NO	PRI	NULL	auto_increment
assessment_id	int	NO	MUL	NULL	
question_id	int	NO	MUL	NULL	
score	int	YES		NULL	

Table 2.3: Questions

- **Purpose:** It contains the questions to be answered by the user.
- **Primary Key:** id(Integer)
- **Data Types and Constraints:**

Field	Type	Null	Key	Default	Extra
question_id	INT	NO	PRI	NULL	auto_increment
question_text	TEXT	NO		NULL	
max_score	INT	YES		**20**	

Table 2.4: Assessments

- **Purpose:** To calculate the score of the user.
- **Primary Key:** id(Integer)
- **Data Types and Constraints:**

Field	Type	Null	Key	Default	Extra
assessment_id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	MUL	NULL	
submission_date	timestamp	YES		CURRENT	DEFAULT
total_score	int	YES		NULL	

Table 2.5: Mental Health Reference

- **Purpose:** It stores the reference values of the questions.
- **Primary Key:** id(Integer)
- **Data Types and Constraints:**

Field	Type	Null	Key	Default	Extra
ref_id	int	NO	PRI	NULL	auto_increment
score_range_min	int	NO		NULL	
score_range_max	int	NO		NULL	
status_category	varchar(100)	NO		NULL	
description	text	YES		NULL	
recommendations	text	YES		NULL	
severity_level	varchar(50)	YES		NULL	

2.3 Creation of Database and Tables - DDL Commands

- **CREATE DATABASE:**

```
CREATE DATABASE mental_health_tracker;
```

- **CREATE USER TABLE:**

```
CREATE TABLE users (
 user_id INT PRIMARY KEY AUTO_INCREMENT,
 user_name VARCHAR(100) NOT NULL,
user_email VARCHAR(100) UNIQUE NOT NULL,
 created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

- **CREATE QUESTIONS TABLE:**

```
CREATE TABLE questions (
question_id INT PRIMARY KEY AUTO_INCREMENT,
question_text TEXT NOT NULL,
```

```
max_score INT DEFAULT 20
);
```

- **CREATE ASSESSMENT TABLE:**

```
CREATE TABLE assessments (
assessment_id INT PRIMARY KEY AUTO_INCREMENT,
user_id INT NOT NULL,
submission_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
 total_score INT,
FOREIGN KEY (user_id)
REFERENCES users(user_id)
ON DELETE CASCADE
);
```

- **CREATE RESPONSE TABLE:**

```
CREATE TABLE responses (
response_id INT PRIMARY KEY AUTO_INCREMENT,
assessment_id INT NOT NULL,
question_id INT NOT NULL,
score INT CHECK (score BETWEEN 1 AND 20),
FOREIGN KEY (assessment_id)
REFERENCES assessments(assessment_id)
ON DELETE CASCADE,
FOREIGN KEY (question_id)
REFERENCES questions(question_id)
ON DELETE CASCADE
```

```
);
```

- **CREATE REFERENCE TABLE:**

```
CREATE TABLE mental_health_reference (
ref_id INT PRIMARY KEY AUTO_INCREMENT,
score_range_min INT NOT NULL,
score_range_max INT NOT NULL,
status_category VARCHAR(100) NOT NULL,
status_description TEXT,
recommendations TEXT,
severity_level VARCHAR(50),
CHECK (score_range_min <= score_range_max)
);
```

2.4 Insertion of tuples into the table – DML commands

- **INSERT INTO REFERENCE:**

```
INSERT INTO mental_health_reference
(score_range_min, score_range_max, status_category, status_description,
recommendations, severity_level)
VALUES
(10, 60, 'Stable',
'You show minimal signs of stress or emotional distress.',
'Maintain healthy routines, exercise regularly, and continue positive habits.',
'Low'),
(61, 110, 'Mild Stress',
```

'You may be experiencing mild stress or emotional imbalance.',

'Practice mindfulness, improve sleep cycle, and monitor emotional changes.',

'Moderate');

- **INSERT INTO QUESTIONS:**

INSERT INTO questions (question_text) VALUES

('How often do you feel overwhelmed?'),('How often do you feel anxious?'),('Do you have trouble sleeping?'),('Do you feel emotionally drained?'),('Do you feel lack of motivation?'),('Do you feel socially withdrawn?'),('Do you experience mood swings?'),('Do you feel difficulty concentrating?'),('Do you feel stressed frequently?'),('Do you feel mentally exhausted?');

- **INSERT INTO USER:**

INSERT INTO users (user_name, user_email) VALUES

('Rahul Sharma', 'rahul@example.com'),

('Ananya Iyer', 'ananya@example.com');

- **INSERT INTO ASSESSMENT:**

INSERT INTO assessments (user_id, total_score) VALUES

(1, 135),

(2, 72);

- **INSERT INTO RESPONSE:**

INSERT INTO responses (assessment_id, question_id, score) VALUES

(1,1,15),(1,2,14),(1,3,13),(1,4,16),(1,5,12),

(1,6,14),(1,7,13),(1,8,15),(1,9,11),(1,10,12);

CHAPTER 3

Complex queries based on the concepts of constraints, sets, joins, views, Triggers and Cursors.

3.1 Adding Constraints and queries based on constraints

Question 1: Ensure that invalid scores (outside 1–20) are not inserted into the responses table.

SQL Statement: INSERT INTO responses (assessment_id, question_id, score) VALUES (1, 1, 25);

Output:

#	Time	Action
1	15:29:21	INSERT INTO responses (assessment_id, question_id, score) VALUES (1, 1, 25)

Figure 3.1.1: Error message indicating violation of CHECK constraint.

Question 2: Insert a valid response into the responses table.

SQL Statement: INSERT INTO responses (assessment_id, question_id, score) VALUES (1, 2, 15);

Output:

8	15:35:24	INSERT INTO responses (assessment_id, question_id, score) VALUES (1, 2, 15)

Figure 3.1.2: Record inserted successfully.

Question 3: Ensure that duplicate emails are not allowed in users table.

SQL Statement: INSERT INTO users (user_name, user_email)

VALUES ('Test User', 'rahul@example.com');

Output:

Error Code: 1062. Duplicate entry 'rahul@example.com' for key 'users.user_email'

Figure 3.1.3: Error due to UNIQUE constraint violation.

3.2 Queries based on Aggregate Functions

Question 1: Find the average score of all assessments.

SQL Statement: SELECT AVG(total_score) AS average_score FROM assessments;

Output:

Result Grid | Filter

	average_score
▶	114.0000

Figure 3.2.1: Displays average total score

Question 2: Find the highest score achieved.

SQL Statement: SELECT MAX(total_score) AS max_score FROM assessments;

Output:

	max_score
▶	114

Figure 3.2.2: Displays highest score.

Question 3: Find the lowest score achieved.

SQL Statement: SELECT MIN(total_score) AS min_score FROM assessments;

Output:

Result Grid | Filter Rows:

min_score
114

Figure 3.2.3: Displays lowest total score.

3.3 Complex queries based on Sets

Question 1: Retrieve all user IDs from both users and assessments tables without duplicates.

SQL Statement: SELECT user_id FROM users UNION
SELECT user_id FROM assessments;

Output:

Result Grid

user_id
2
1

Figure 3.3.1: List of unique user IDs

Question 2: Retrieve all user IDs including duplicates.

SQL Statement: SELECT user_id FROM users UNION ALL SELECT user_id FROM assessments;

Output:

Result Grid

user_id
2
1
1
2

Figure 3.3.2: List of user IDs including duplicates.

Question 3: Find common user IDs using logical comparison.

SQL Statement: SELECT user_id FROM users WHERE user_id IN (SELECT user_id FROM assessments);

Output:

Result Grid

user_id
2
1
NULL

Figure 3.3.3: List of user IDs present in both tables.

3.4 Complex queries based on Subqueries

Question 1: Find assessments with scores above average.

SQL Statement: SELECT * FROM assessments WHERE total_score > (SELECT AVG(total_score) FROM assessments);

Output:

Result Grid Filter Rows: Edit:

assessment_id	user_id	submission_date	total_score
NULL	NULL	NULL	NULL

Figure 3.4.1: Displays assessments with above-average scores.

Question 2: Find users who have taken assessments.

SQL Statement: SELECT user_name FROM users WHERE user_id IN (SELECT user_id FROM assessments);

Output:

Result Grid

user_name
Rahul Sharma
Ananya Iyer

Figure 3.4.2: List of users who attempted the test.

Question 3: Find the assessment with the highest score.

SQL Statement: SELECT * FROM assessments WHERE total_score = (SELECT MAX(total_score) FROM assessments);

Output:

Result Grid Filter Rows: Edit:

assessment_id	user_id	submission_date	total_score
1	1	2026-03-21 08:29:24	124
NULL	NULL	NULL	NULL

Figure 3.4.3: Displays the highest scoring assessment.

3.5 Complex queries based on Joins

Question 1: Display user names with their total scores.

SQL Statement: SELECT u.user_name, a.total_score FROM users u JOIN assessments a ON u.user_id = a.user_id;

Output:

Result Grid | Filter Rows:

user_name	total_score
Rahul Sharma	124
Ananya Iyer	NULL

Figure 3.5.1: Shows user names and scores.

Question 2: Display user responses with question text.

SQL Statement: SELECT u.user_name, q.question_text, r.score FROM users u
JOIN assessments a ON u.user_id = a.user_id
JOIN responses r ON a.assessment_id = r.assessment_id
JOIN questions q ON r.question_id = q.question_id;

Output:

Result Grid | Filter Rows: | Export:

user_name	question_text	score
Rahul Sharma	Do you feel stressed frequently?	10
Rahul Sharma	Do you have trouble sleeping?	12
Rahul Sharma	Do you feel anxious without reason?	11
Rahul Sharma	Do you feel emotionally drained?	13
Rahul Sharma	Do you lose interest in activities?	9
Rahul Sharma	Do you feel irritated often?	10
Rahul Sharma	Do you struggle to concentrate?	11
Rahul Sharma	Do you feel hopeless?	12
Rahul Sharma	Do you experience mood swings?	10
Rahul Sharma	Do you avoid social interaction?	11
Rahul Sharma	Do you have trouble sleeping?	15
Ananya Iyer	Do you feel stressed frequently?	5
Ananya Iyer	Do you have trouble sleeping?	6
Ananya Iyer	Do you feel anxious without reason?	7
Ananya Iyer	Do you feel emotionally drained?	6
Ananya Iyer	Do you lose interest in activities?	5
Ananya Iyer	Do you feel irritated often?	6

Figure 3.5.2: Displays user responses for each question.

Question 3: Display assessment results with mental health category.

SQL Statement: SELECT a.assessment_id, a.total_score, m.status_category FROM assessments a
JOIN mental_health_reference m ON a.total_score BETWEEN m.score_range_min AND m.score_range_max;

Output:

	assessment_id	total_score	status_category
▶	1	124	Moderate Distress

Figure 3.5.3: Displays classification of each assessment.

3.6 Complex queries based on views

Question 1: Create a view for user scores.

SQL Statement: CREATE VIEW user_scores AS
SELECT u.user_name, a.total_score FROM users u
JOIN assessments a ON u.user_id = a.user_id;

Output: View created successfully.

Question 2: Retrieve data from the view.

SQL Statement: SELECT * FROM user_scores;

Output:

	user_name	total_score
▶	Rahul Sharma	124
	Ananya Iyer	NULL

Figure 3.6.1: Displays user names and scores.

Question 3: Create a view for assessment classification.

SQL Statement: CREATE VIEW assessment_results AS
SELECT a.assessment_id, m.status_category FROM assessments a
JOIN mental_health_reference m ON a.total_score BETWEEN m.score_range_min AND m.score_range_max;

Output: View created successfully.

3.7 Complex queries based on Triggers

Question 1: Automatically update total score when a response is inserted.

SQL Statement:

```
DELIMITER $$
CREATE TRIGGER calculate_total_score
AFTER INSERT ON responses
FOR EACH ROW
BEGIN
UPDATE assessments
SET total_score = ( SELECT SUM(score) FROM responses WHERE assessment_id
= NEW.assessment_id)
WHERE assessment_id = NEW.assessment_id;
END$$
DELIMITER ;
```

Output: Trigger created successfully.

Question 2: Insert response to test trigger.

SQL Statement: INSERT INTO responses (assessment_id, question_id, score) VALUES (1, 3, 12);

Output: Response inserted and total_score updated automatically.

Question 3: Check updated total score.

SQL Statement: SELECT total_score FROM assessments WHERE assessment_id = 1;

Output:

Result Grid

	total_score
▶	136

Figure 3.7.1: Updated total score displayed.

3.8 Complex queries based on Cursors

Question 1: Retrieve all user names one by one using a cursor.

SQL Statement:

```
DELIMITER $$
CREATE PROCEDURE get_all_users()
BEGIN
  DECLARE done INT DEFAULT FALSE;
  DECLARE uname VARCHAR(100);
  DECLARE cur CURSOR FOR
    SELECT user_name FROM users;
  DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
  OPEN cur;
  read_loop: LOOP
    FETCH cur INTO uname;
    IF done THEN
      LEAVE read_loop;
    END IF;
    SELECT uname AS user_name;
  END LOOP;
```

```
  CLOSE cur;
END$$
DELIMITER ;
CALL get_all_users();
```

Output:

Result Grid | Filter R

	user_name
▶	Ananya Iyer

Figure 3.8.1: List of user names displayed one by one.

Question 2: Retrieve all assessment scores using a cursor.

SQL Statement: CALL get_all_scores();

Output: All assessment scores displayed sequentially.

Question 3: Retrieve user names along with their scores using a cursor.

SQL Statement: CALL get_user_scores_cursor();

Output: Displays user names along with their total scores one by one.

CHAPTER 4
ANALYZING THE PITFALLS, IDENTIFYING THE DEPENDENCIES, AND APPLYING NORMALIZATIONS

4.1 Analyse the Pitfalls in Relations

Initially, the database may contain redundancy and anomalies due to improper design.

Common Pitfalls:

- Data Redundancy → Same data repeated (e.g.,username repeated in assessment table)
- Update Anomaly → Updating data in multiple places
- Insertion Anomaly → Cannot insert data without other data
- Deletion Anomaly → Losing important data when deleting

Example:
If the user name is stored in the assessment table, updating the name requires multiple updates.

4.2 First Normal Form

4.2.1: Identify Dependency

Each attribute must contain atomic values

No repeating groups

Before 1F:

Response_id	Assessment_id	Question_id	Score
1	101	1,2,3	5,3,4

4.2.2: Apply Normalization to 1NF

Response_id	Assessment_id	Question_id	Score
1	101	1	5
1	101	2	3
1	101	3	4

4.3 Second Normal Form

4.3.1: Identify Dependency

- Remove partial dependency
- Non-key attributes depend on full primary key

BEFORE 2NF:

Response_id	Assessment_id	Question_id	Score
1	101	1	5
1	101	2	3
1	101	3	4

4.3.2: Apply Normalization to 2NF

RESPONSE TABLE

Response_id	Assessment_id	Question_id	Score
1	101	1	5
2	101	2	3
3	101	3	4

QUESTION TABLE

Question_id	question_text
1	Stress
2	Sleep
3	Anxiety

4.4 Third Normal Form

4.4.1: Identify Dependency

- Remove transitive dependency

RESPONSE TABLE

Response_id	Assessment_id	Question_id	Score
1	101	1	5
2	101	2	3
3	101	3	4

QUESTION TABLE

Question_id	question_text
1	Stress
2	Sleep
3	Anxiety

4.4.2: Apply Normalization to 3NF

ASSESSMENT TABLE

Assessment_id	Total_Score
101	12

REFERENCE TABLE

score_min	score_max	category
10	60	Stable

4.5 BCNF

4.5.1: Identify Dependency

- Every determinant must be a candidate key

4.5.2: Apply Normalization to BCNF

- The responses table satisfies BCNF as every functional dependency has a determinant that is a candidate key. Therefore, no further decomposition is required and the table is free from redundancy and anomalies.

4.6 Fourth Normal Form

4.6.1: Identify Dependency

- Each row represents **one question response**
- No attribute contains multiple independent values

4.6.2: Apply Normalization to 4NF

- The table satisfies Fourth Normal Form as it does not contain any multivalued dependencies. Each record represents a single fact, ensuring no redundancy due to independent multi-valued attributes.

4.7 Fifth Normal Form

4.7.1: Identify Dependency

- responses cannot be further decomposed meaningfully
- Splitting would lose logical relationship between:
    - assessment
    - question
    - score

4.7.2: Apply Normalization to 5NF

The table is already in 5NF as:

- No further decomposition is possible
- No redundancy due to join dependencies

- The table satisfies Fifth Normal Form as there are no join dependencies and it cannot be further decomposed without loss of information. Hence, the current structure is optimal.

CHAPTER 5
IMPLEMENTATION OF CONCURRENCY CONTROL AND RECOVERY MECHANISMS

5.1 Introduction to Transactions

A transaction is a sequence of SQL operations executed as a single unit.

5.1.1 Properties

- Atomicity → All or nothing
- Consistency → Database remains valid
- Isolation → Transactions don't interfere
- Durability → Changes are permanent

5.1.2 States

- Active
- Partially Committed
- Committed
- Failed
- Aborted

5.2 Transaction Control Language (TCL)

5.2.1 Save point

Temporary point in transaction

SAVEPOINT sp1;

5.2.2 Commit

Saves changes permanently

COMMIT;

5.2.3 Rollback

Undo changes

ROLLBACK;

5.3 create 5 transactions for your project and execute

5.3.1 Transaction1 code with output (min 5 transactions)

Use savepoints, commit, rollback

< Sample:

```
-- Step 1: Start transaction by updating question value
UPDATE response
SET score = score + 5 WHERE question_id = 3;

-- Step 2: Set a savepoint after first change
SAVEPOINT after_first_update;

-- Step 3: Another update
UPDATE response SET score = score – 5 WHERE question_id = 2;

-- Step 4: Oops! Something's wrong with second update, rollback to savepoint
ROLLBACK TO after_first_update;

-- Step 5: Continue with a safe update
UPDATE response SET score = score + 5 WHERE question_id = 3;


-- Step 6: Commit all changes since savepoint
COMMIT; >
```

5.3 Concurrency control

5.3.1 Concurrency control Algorithms

5.3.1 Locking commands

a. Row-Level Locking – SELECT ... FOR UPDATE

b. Table-Level Locking – LOCK TABLE

Lock Modes

Lock Mode	Description
ROW SHARE	Allows concurrent access; prevents other sessions from locking the table exclusively.
ROW EXCLUSIVE	Prevents other sessions from locking in share mode. Used by default for DML.
SHARE	Allows queries but not updates or deletes.
SHARE ROW EXCLUSIVE	A mix; more restrictive than SHARE.
EXCLUSIVE	Prevents all other access — full table lock.

c. COMMIT – Release All Locks

d. ROLLBACK – Undo Changes & Release Locks

5.3.2 Example (1 example for your project)

```
START TRANSACTION;

SELECT * FROM assessments
WHERE assessment_id = 1
FOR UPDATE;

UPDATE assessments
SET total_score = total_score + 10
WHERE assessment_id = 1;

COMMIT;
```

CHAPTER 6

FRONT-END AND BACK-END CODE OF Mental Health Assessment and Analysis System

6.1 Front–End Module codes

6.1.1 Index Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>MindCheck — Mental Health Analyser</title>
  <link rel="preconnect" href="https://fonts.googleapis.com" />
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
<link
href="https://fonts.googleapis.com/css2?family=Lora:ital,wght@0,400;0,500;0,600;1,400&fa
mily=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet" />
  <link rel="stylesheet" href="/static/css/style.css" />
  <!-- If opening directly in browser (no server), use the line below instead and comment out
the one above -->
  <!-- <link rel="stylesheet" href="style.css" /> -->
</head>
<body>


  <!-- NAV -->
  <nav id="navbar">
```

```
  <a href="index.html" class="nav-logo">
    <span class="dot"></span>MindCheck
  </a>
  <ul class="nav-links">
    <li class="hide-mobile"><a href="#how">How it works</a></li>
    <li><a href="/questions" class="nav-cta">Take the test</a></li>
  </ul>
</nav>


<!-- HERO -->
<section class="hero">
  <div class="hero-text">
    <div class="hero-badge"><span></span>Academic Research Project</div>
    <h1 class="hero-title">
      How are you <em>really</em><br>feeling today?
    </h1>
    <p class="hero-subtitle">
        Answer 10 simple questions about your daily well-being. Our system analyses your responses and provides a personalised mental health snapshot — in under 3 minutes.
    </p>
    <div class="hero-actions">
      <a href="/questions" class="btn-primary">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 5l7 7-7 7"/></svg>
        Start assessment
      </a>
      <a href="#how" class="btn-ghost">Learn how it works</a>
```

```
    </div>
    <div class="hero-trust">
      <div class="trust-item">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 22s8-4 8-10V5l-8-3-8 3v7c0 6 8 10 8 10z"/></svg>
        Anonymous &amp; secure
      </div>
      <div class="trust-item">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle cx="12" cy="12" r="10"/><polyline points="12 6 12 12 16 14"/></svg>
        Under 3 minutes
      </div>
      <div class="trust-item">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 12l2 2 4-4m6 2a9 9 0 11-18 0 9 9 0 0118 0z"/></svg>
        Not a medical tool
      </div>
    </div>
  </div>



 </section>


 <!-- HOW IT WORKS -->
 <section class="section" id="how">
  <div class="section-label">Process</div>
```

```
  <h2 class="section-title">Three steps to clarity</h2>

    <p class="section-sub">No sign-ups. No complexity. Just honest answers and a clear picture of where you stand.</p>


  <div class="how-grid">

    <div class="how-card">

      <div class="how-number">01</div>

      <div class="how-chip chip-terra">

        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M11 4H4a2 2 0 00-2 2v14a2 2 0 002 2h14a2 2 0 002-2v-7"/><path d="M18.5 2.5a2.121 2.121 0 013 3L12 15l-4 1 1-4 9.5-9.5z"/></svg>

      </div>

      <h3>Answer 10 questions</h3>

        <p>Each question explores a different dimension of your mental well-being — stress, sleep, mood, and more. Simple, honest options. No right or wrong answers.</p>

    </div>


    <div class="how-card">

      <div class="how-number">02</div>

      <div class="how-chip chip-sage">

                <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle cx="12" cy="12" r="3"/><path d="M12 1v4M12 19v4M4.22 4.22l2.83 2.83M16.95 16.95l2.83 2.83M1 12h4M19 12h4M4.22 19.78l2.83-2.83M16.95 7.05l2.83-2.83"/></svg>

      </div>

      <h3>We analyse your responses</h3>

        <p>Your answers are processed by our system. Each response is weighed against established well-being indicators to build an accurate picture.</p>
```

```
    </div>


    <div class="how-card">
      <div class="how-number">03</div>
      <div class="how-chip chip-warm">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 00-2 2v16a2 2 0 002 2h12a2 2 0 002-2V8z"/><polyline points="14 2 14 8 20 8"/><line x1="16" y1="13" x2="8" y2="13"/><line x1="16" y1="17" x2="8" y2="17"/></svg>
      </div>
      <h3>See your result</h3>
        <p>You receive a clear mental health snapshot with context — what it means for you and what steps you can consider going forward.</p>
    </div>
  </div>
 </section>


 <!-- ANSWER HONESTLY -->
 <section class="section" style="max-width: 640px; margin: 0 auto; padding-top: 0;">
  <div class="honesty-card">
    <div class="honesty-icon">
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M20.84 4.61a5.5 5.5 0 00-7.78 0L12 5.67l-1.06-1.06a5.5 5.5 0 00-7.78 7.78l1.06 1.06L12 21.23l7.78-7.78 1.06-1.06a5.5 5.5 0 000-7.78z"/></svg>
    </div>
    <h3>Answer honestly</h3>
```

```
      <p>This tool works best when you reflect genuinely. Your responses are private and used only to generate your personal result. There is nothing to gain from choosing any particular answer.</p>

     <ul class="honesty-list">

       <li>

         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 12l2 2 4-4"/><circle cx="12" cy="12" r="10"/></svg>

         Take your time with each question

       </li>

       <li>

         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 12l2 2 4-4"/><circle cx="12" cy="12" r="10"/></svg>

         Go with your first instinct

       </li>

       <li>

         <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 12l2 2 4-4"/><circle cx="12" cy="12" r="10"/></svg>

         Think about the past two weeks

       </li>

     </ul>

   </div>

   <div class="disclaimer" style="margin-top: 1.25rem;">

       <strong>Academic project notice:</strong> MindCheck is built as a DBMS academic project. It is not a clinical tool and does not replace professional mental health evaluation. If you are experiencing distress, please consult a qualified mental health professional.

   </div>

 </section>
```

```
<!-- CTA STRIP -->
<section class="cta-strip">
  <h2>Ready to check in with <em>yourself?</em></h2>
  <p>It takes 3 minutes. No account needed. Your responses stay private.</p>
  <a href="/questions" class="btn-primary btn-large">
      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path
d="M9 5l7 7-7 7"/></svg>
    Begin assessment
  </a>
</section>

<!-- FOOTER -->
<footer>
  <span class="footer-logo">MindCheck</span>
  <p>Built as a DBMS project &mdash; <a
href="https://github.com/Dia-Garg/Mental_health_analyser" target="_blank">View on
GitHub</a></p>
  <p>Not a medical diagnostic tool &mdash; for academic use only</p>
</footer>

<script src="/static/js/main.js"></script>
</body>
</html>
```

6.1.2 Questionnaire Page

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Assessment — MindCheck</title>

  <link rel="preconnect" href="https://fonts.googleapis.com" />

  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />

                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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href="https://fonts.googleapis.com/css2?family=Lora:ital,wght@0,400;0,500;0,600;1,400&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet" />

  <link rel="stylesheet" href="/static/css/style.css" />

</head>

<body>


  <!-- NAV -->

  <nav id="navbar">

    <a href="/" class="nav-logo">

      <span class="dot"></span>MindCheck

    </a>

    <ul class="nav-links">

      <li><a href="/">← Back to home</a></li>

    </ul>

  </nav>


  <!-- QUESTIONNAIRE -->
```

```
<main class="quiz-page">


  <div class="quiz-header">
    <h1>Mental Health Assessment</h1>
    <p>Answer each question honestly based on how you have been feeling over the past two weeks.</p>
  </div>


  <!-- PROGRESS BAR -->
  <div class="quiz-progress-wrap">
    <div class="quiz-progress-info">
      <span id="progress-label">Question 1 of 11</span>
      <span id="progress-percent">0% complete</span>
    </div>
    <div class="quiz-progress-bar">
      <div class="quiz-progress-fill" id="quiz-progress-fill"></div>
    </div>
  </div>


  <!-- STEP 0: USER INFO -->
  <div class="quiz-step" id="step-0">
    <div class="user-info-section">
      <h2>Before we begin</h2>
      <div class="form-group">
        <label for="user-name">Your name</label>
                <input type="text" id="user-name" placeholder="e.g. Ananya Iyer" autocomplete="off" />
```

```
      </div>
      <div class="form-group">
        <label for="user-email">Email address</label>
                    <input type="email" id="user-email" required placeholder="e.g. ananya@example.com" autocomplete="off" />
      </div>
        <div id="info-error" class="field-error" style="display:none;">Please fill in both fields to continue.
      {% if error %}
        <div class="field-error" style="display:block;">
          {{ error }}
        </div>
      {% endif %}
      </div>


    </div>
    <div style="text-align:center;">
      <button class="btn-primary" onclick="startQuiz()">
                    <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 5l7 7-7 7"/></svg>
        Begin assessment
      </button>
    </div>
  </div>


  <!-- STEPS 1–10: QUESTIONS -->
  <div id="questions-container"></div>
```

```
  <!-- SUBMIT -->
  <div class="quiz-submit" id="submit-section" style="display:none;">
        <div class="disclaimer" style="max-width:520px; margin: 0 auto 2rem; text-align:left;">
      <strong>Before you submit:</strong> Make sure all your answers reflect how you have genuinely been feeling. Your result is based entirely on your responses.
    </div>
    <button class="btn-primary btn-large" onclick="submitQuiz()">
                <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M9 12l2 2 4-4m6 2a9 9 0 11-18 0 9 9 0 0118 0z"/></svg>
      See my result
    </button>
  </div>

 </main>

 <!-- FOOTER -->
 <footer>
  <span class="footer-logo">MindCheck</span>
                <p>Built as a DBMS project &mdash; <a href="https://github.com/Dia-Garg/Mental_health_analyser" target="_blank">View on GitHub</a></p>
  <p>Not a medical diagnostic tool &mdash; for academic use only</p>
 </footer>

 <script src="/static/js/main.js"></script>
 <script src="/static/js/questionnaire.js"></script>
```

```
</body>

</html>
```

6.1.3 Result Page

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Your Result — MindCheck</title>

  <link rel="preconnect" href="https://fonts.googleapis.com" />

  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />

                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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href="https://fonts.googleapis.com/css2?family=Lora:ital,wght@0,400;0,500;0,600;1,40
0&family=DM+Sans:wght@300;400;500&display=swap" rel="stylesheet" />

  <link rel="stylesheet" href="/static/css/style.css" />

</head>

<body>


  <!-- NAV -->

  <nav id="navbar">

    <a href="/" class="nav-logo">

      <span class="dot"></span>MindCheck

    </a>

    <ul class="nav-links">

      <li><a href="/">← Back to home</a></li>

    </ul>
```

```
 </nav>


 <!-- RESULT -->
 <main class="result-page">


  <div class="result-card">
 <p class="result-name">Hi, {{ request.form.user_name if request.form.user_name else
'' }}</p>
 <h2 class="result-category">{{ result }}</h2>


 <div class="result-score-bar-wrap">
  <div class="result-score-bar">
   <div class="result-score-fill" style="width: {{ (score / 200) * 100 }}%"></div>
  </div>
  <span class="result-score-label">Score: {{ score }} / 200</span>
 </div>
</div>


<div class="result-details">
 <h3>Score Interpretation</h3>


 <table class="score-table">
  <thead>
   <tr>
    <th>Score Range</th>
    <th>Category</th>
```

```
      <th>Severity</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>10 – 60</td>
      <td>Stable</td>
      <td>Low</td>
    </tr>
    <tr>
      <td>61 – 110</td>
      <td>Mild Stress</td>
      <td>Moderate</td>
    </tr>
    <tr>
      <td>111 – 150</td>
      <td>Moderate Distress</td>
      <td>High</td>
    </tr>
    <tr>
      <td>151 – 200</td>
      <td>Severe Distress</td>
      <td>Critical</td>
    </tr>
  </tbody>
</table>
```

```
</div>


  <div class="result-actions">
    <a href="/questions" class="btn-primary">
                <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M1 4v6h6"/><path d="M3.51 15a9 9 0 1 0 .49-3.5"/></svg>
      Take it again
    </a>
    <a href="/" class="btn-ghost">Back to home</a>
  </div>


  <div class="disclaimer" style="margin-top: 2rem;">
        <strong>Important:</strong> This result is generated by an academic DBMS project and is not a clinical diagnosis. If you are experiencing significant distress, please speak to a qualified mental health professional.
  </div>


 </main>


 <!-- FOOTER -->
 <footer>
  <span class="footer-logo">MindCheck</span>
                <p>Built as a DBMS project &mdash; <a href="https://github.com/Dia-Garg/Mental_health_analyser" target="_blank">View on GitHub</a></p>
  <p>Not a medical diagnostic tool &mdash; for academic use only</p>
 </footer>
```

```
 <script src="/static/js/main.js"></script>
</body>
</html>
```

6.2 Database connectivity

```
from flask import Flask, render_template, request, redirect
import mysql.connector

app = Flask(__name__)

# DB connection
def get_db_connection():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="Havelock@06",
        database="mental_health_tracker"
    )

# Home page
@app.route('/')
def index():
    return render_template('index.html')
```

```
# Questionnaire page
@app.route('/questions')
def questions():
    return render_template('Questionnaire.html')


# Submit form
@app.route('/submit', methods=['POST'])
def submit():
    print("FORM DATA:", request.form)
    name = request.form['user_name']
    email = request.form['user_email']
    if not name or not email:
        return "Invalid input", 400
    if '@' not in email or '.' not in email:
        return render_template('Questionnaire.html', error="Invalid email format")


    conn = get_db_connection()
    cursor = conn.cursor()


    try:
        cursor.execute(
            "INSERT INTO users (user_name, user_email) VALUES (%s, %s)",
            (name, email)
        )
        user_id = cursor.lastrowid
```

```
except Exception as e:
    print("INSERT ERROR:", e)

    cursor.execute(
        "SELECT user_id FROM users WHERE user_email = %s",
        (email,)
    )
    row = cursor.fetchone()

    if row:
        user_id = row[0]
    else:
        return "User creation failed", 500

# Create assessment
cursor.execute(
    "INSERT INTO assessments (user_id) VALUES (%s)",
    (user_id,)
)
assessment_id = cursor.lastrowid

# Insert responses (assuming 10 questions)
total_score = 0
for i in range(1, 11):
    print(f"q{i}:", request.form.get(f'q{i}'))  # debug
```

```
        score = int(request.form[f'q{i}'])
        total_score += score

    cursor.execute(
        "INSERT INTO responses (assessment_id, question_id, score) VALUES (%s, %s, %s)",
        (assessment_id, i, score)
    )

    conn.commit()

    # Fetch result
    cursor.execute("""
        SELECT status_category
        FROM mental_health_reference
        WHERE %s BETWEEN score_range_min AND score_range_max
    """, (total_score,))

    row = cursor.fetchone()
    if row:
        result = row[0]
    else:
        result = "No category found"

    conn.close()
```

```
    return render_template('result.html', score=total_score, result=result)


@app.route('/test')
def test():
    conn = get_db_connection()
    cursor = conn.cursor()

    # fake user
    cursor.execute(
        "INSERT INTO users (user_name, user_email) VALUES (%s, %s)",
        ("Test User", "testuser123@gmail.com")
    )
    user_id = cursor.lastrowid

    # create assessment
    cursor.execute(
        "INSERT INTO assessments (user_id) VALUES (%s)",
        (user_id,)
    )
    assessment_id = cursor.lastrowid

    total_score = 0

    # simulate 10 responses
    for i in range(1, 11):
        score = 10  # fixed value
```

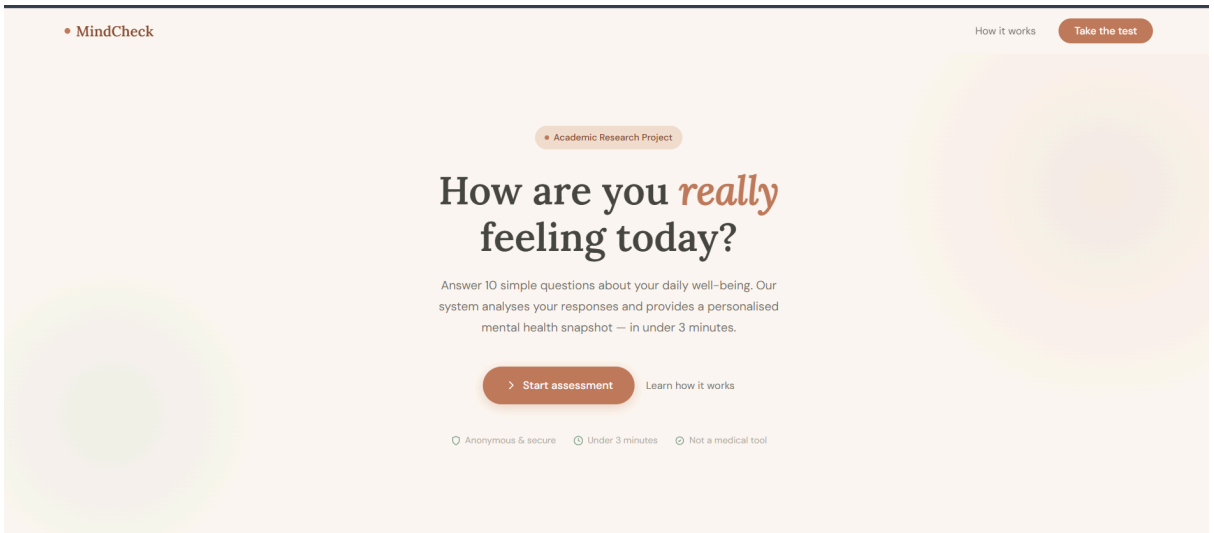
```
        total_score += score

        cursor.execute(
            "INSERT INTO responses (assessment_id, question_id, score) VALUES (%s, %s, %s)",
            (assessment_id, i, score)
        )

    # update total score
    cursor.execute(
        "UPDATE assessments SET total_score = %s WHERE assessment_id = %s",
        (total_score, assessment_id)
    )

    conn.commit()
    conn.close()

    return "TEST INSERT DONE"

if __name__ == '__main__':
    app.run(debug=True)
```

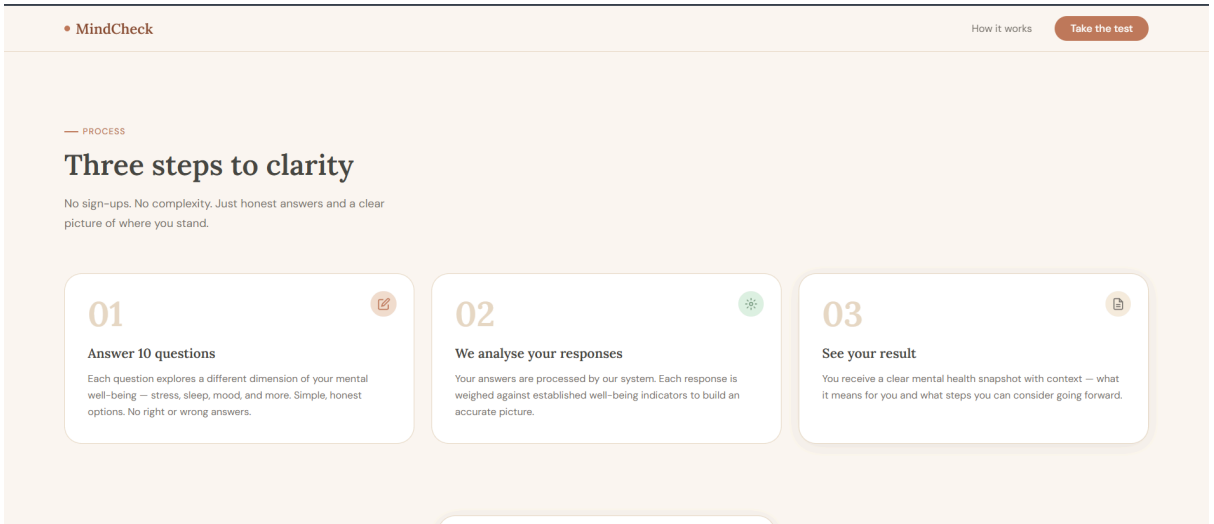
CHAPTER 7

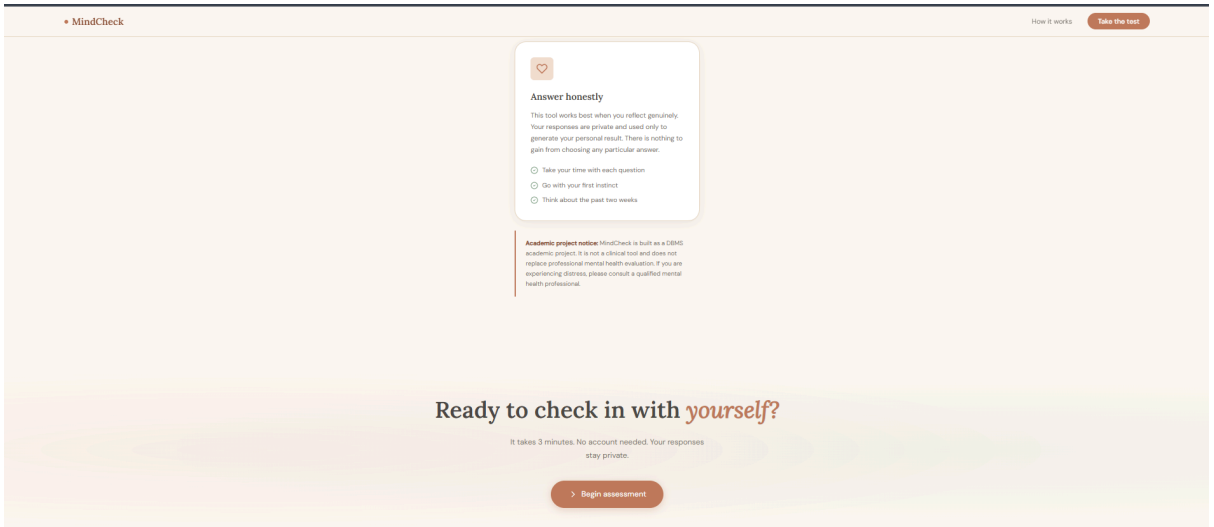
RESULTS AND DISCUSSIONS

7.1 Screenshots of front-end

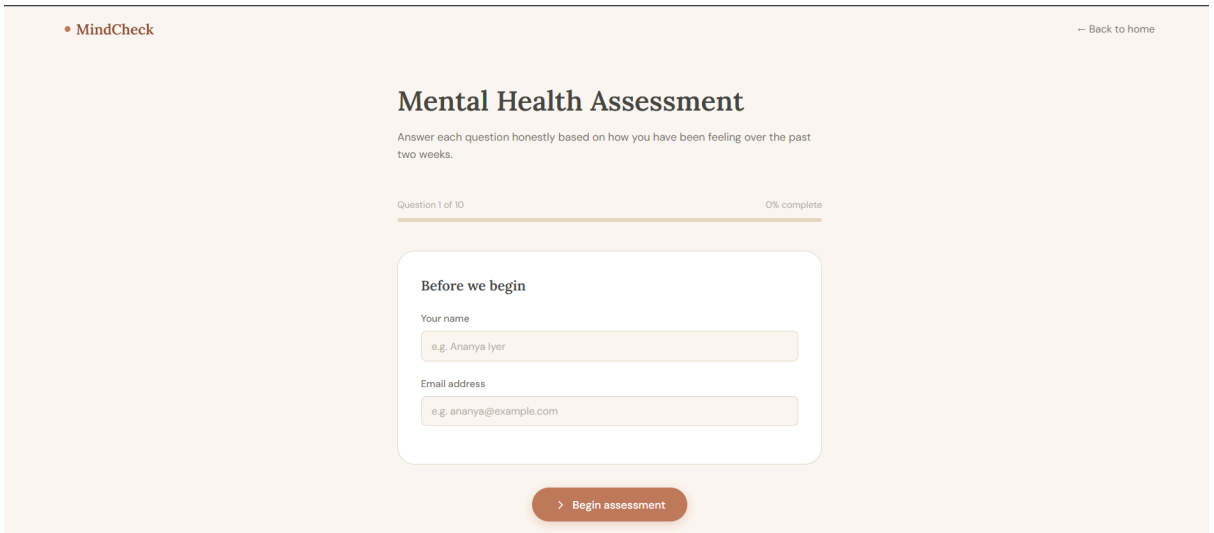
7.1.1 Index Page

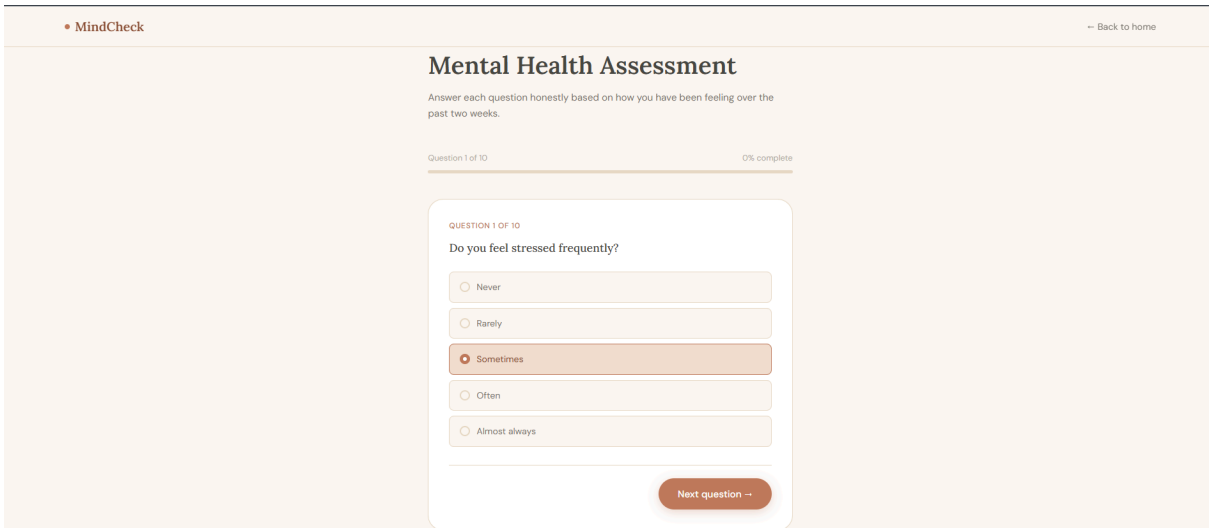


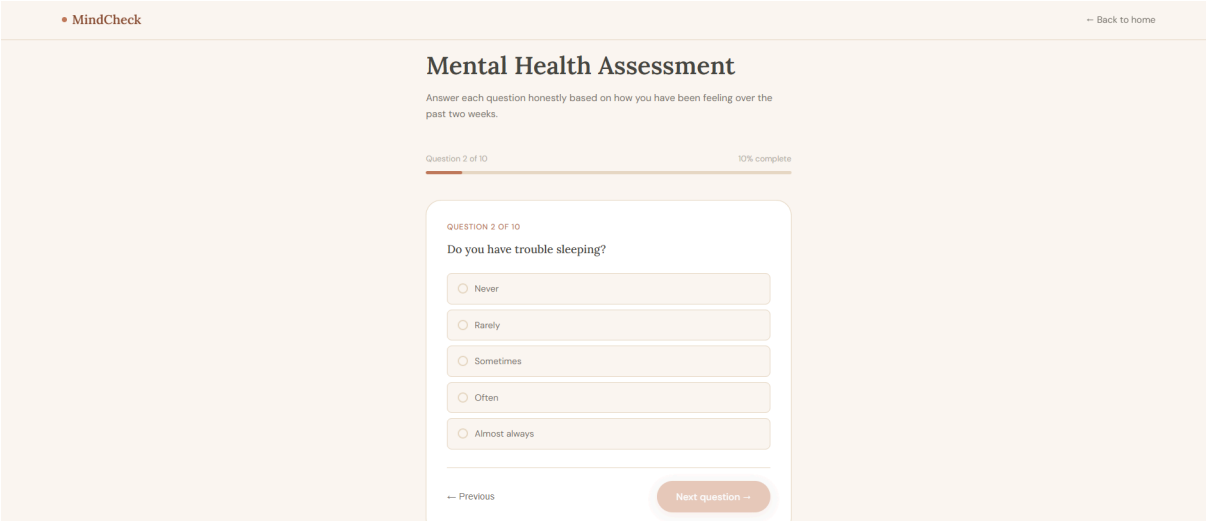


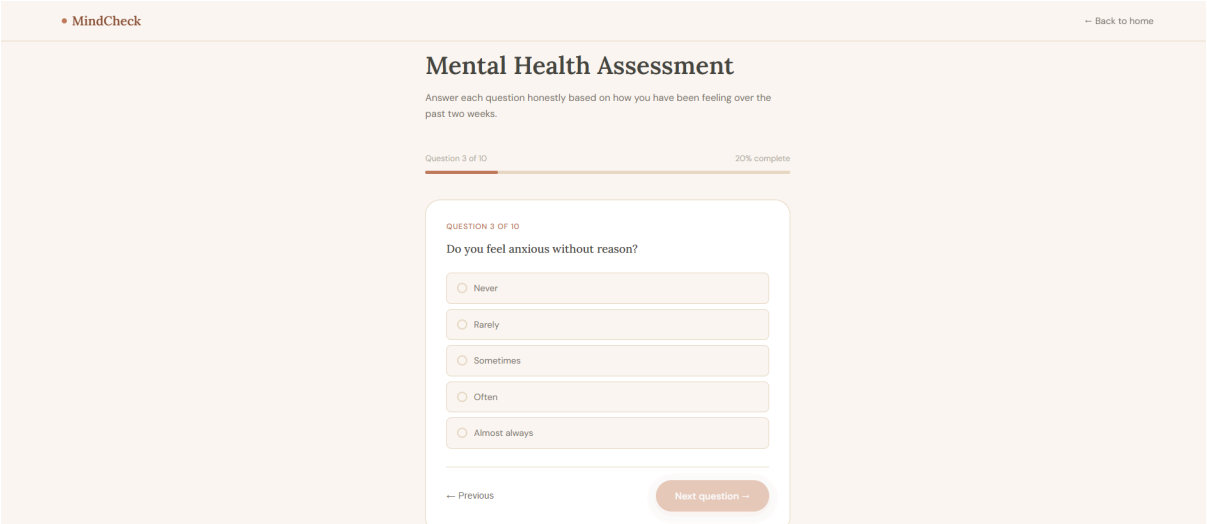


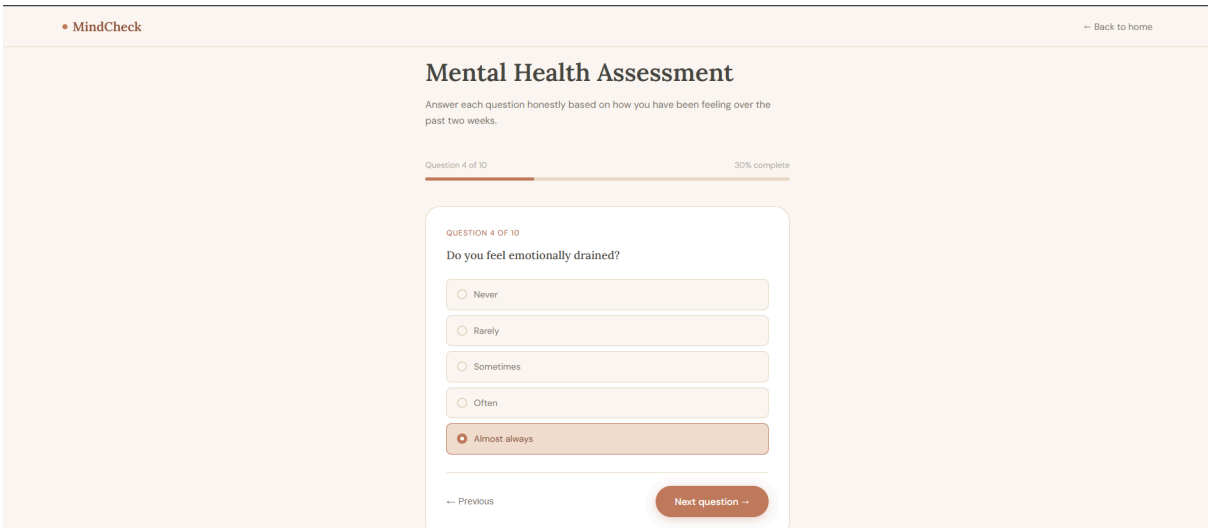
7.1.2 Questionnaire Page

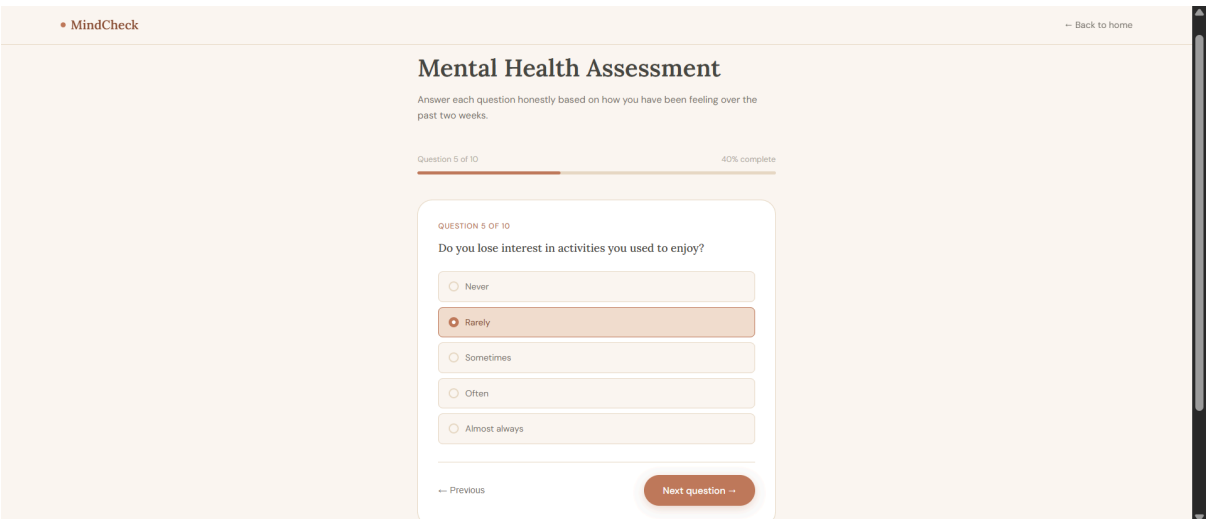


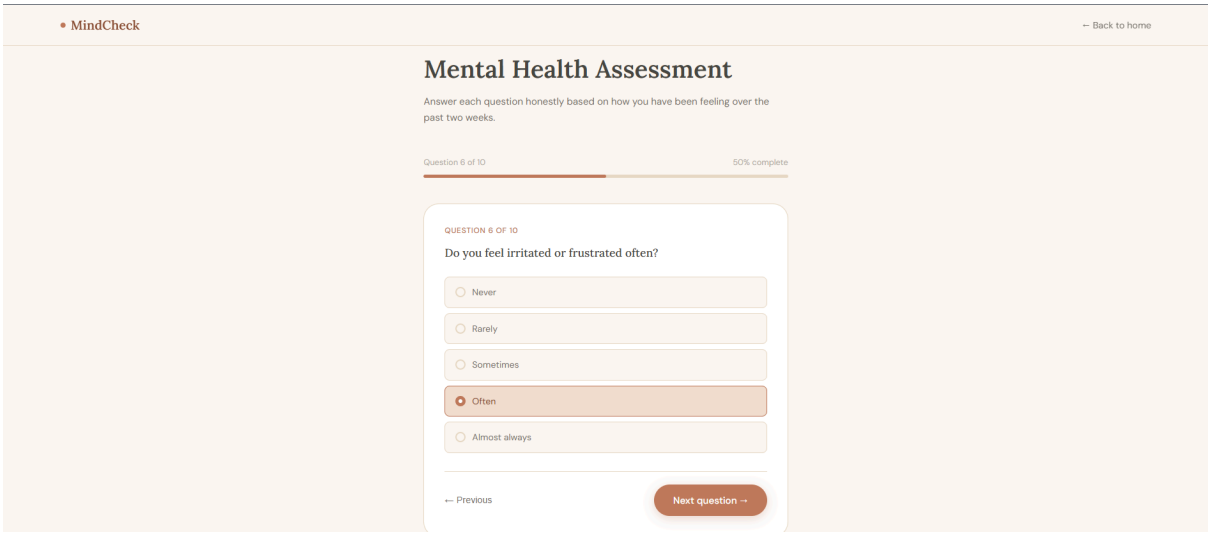


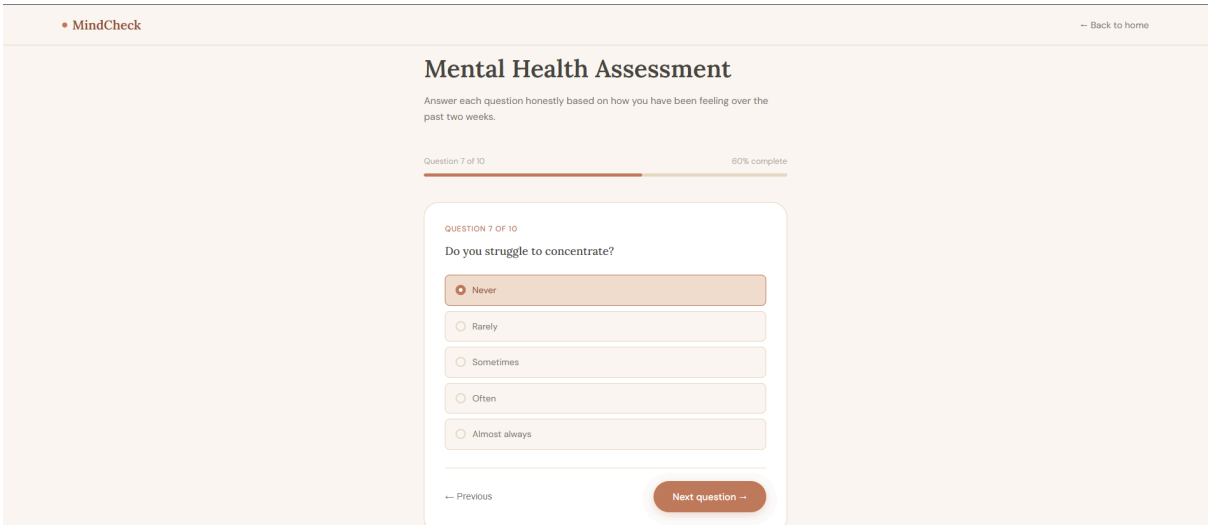

MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 2 of 10
10% complete
QUESTION 2 OF 10
Do you have trouble sleeping?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →

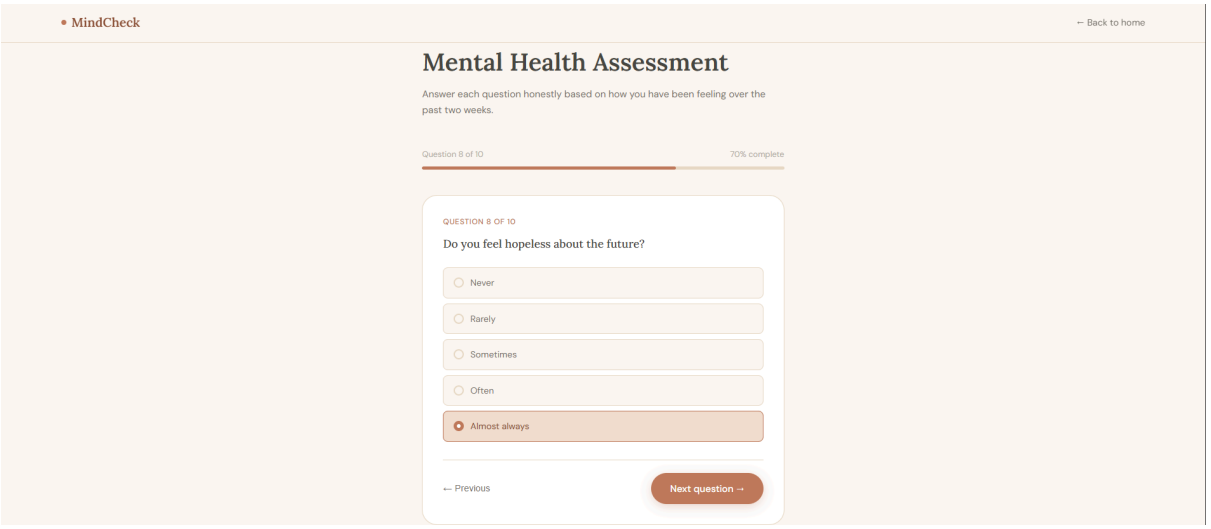

MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 3 of 10
20% complete
QUESTION 3 OF 10
Do you feel anxious without reason?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →

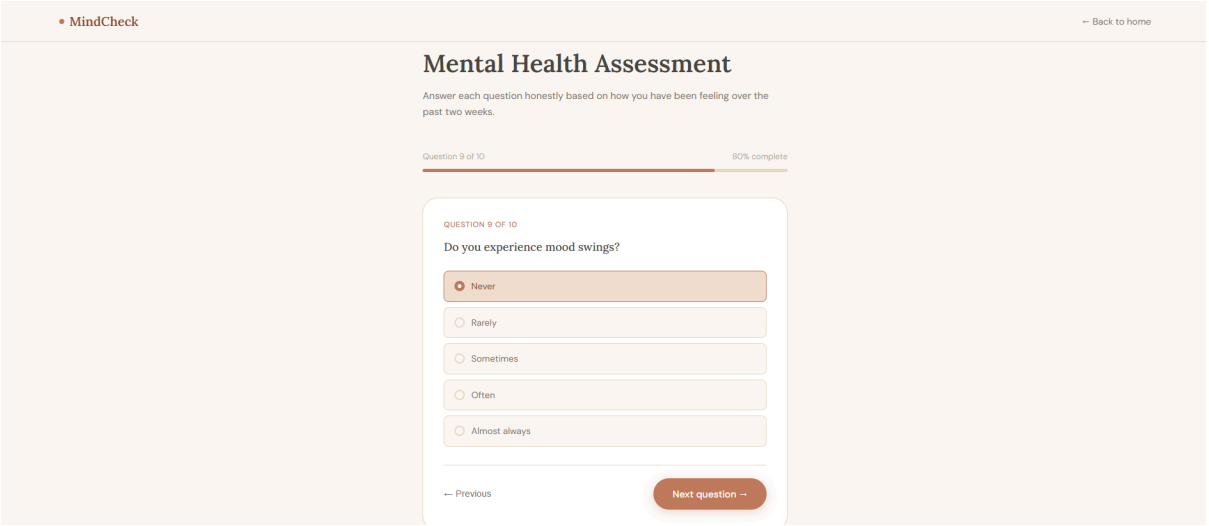

MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 4 of 10
30% complete
QUESTION 4 OF 10
Do you feel emotionally drained?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →

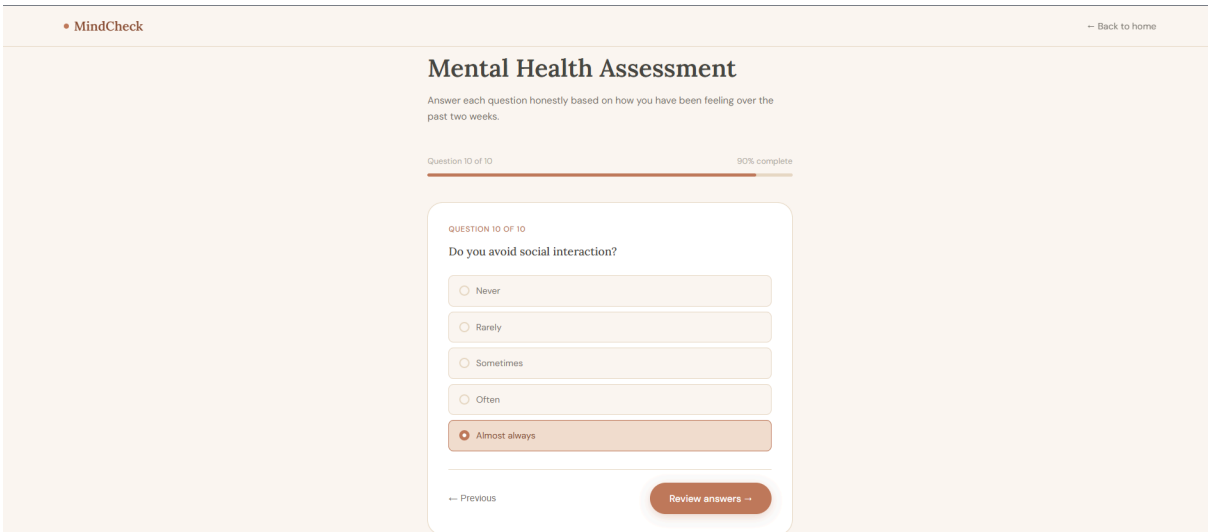

• MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 5 of 10
40% complete
QUESTION 5 OF 10
Do you lose interest in activities you used to enjoy?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →

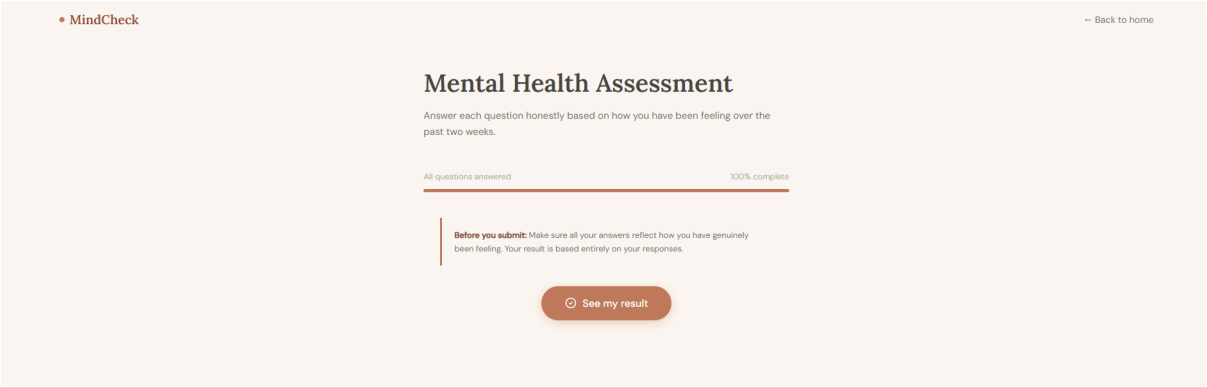

• MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 6 of 10
50% complete
QUESTION 6 OF 10
Do you feel irritated or frustrated often?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →


• MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 7 of 10
60% complete
QUESTION 7 OF 10
Do you struggle to concentrate?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →

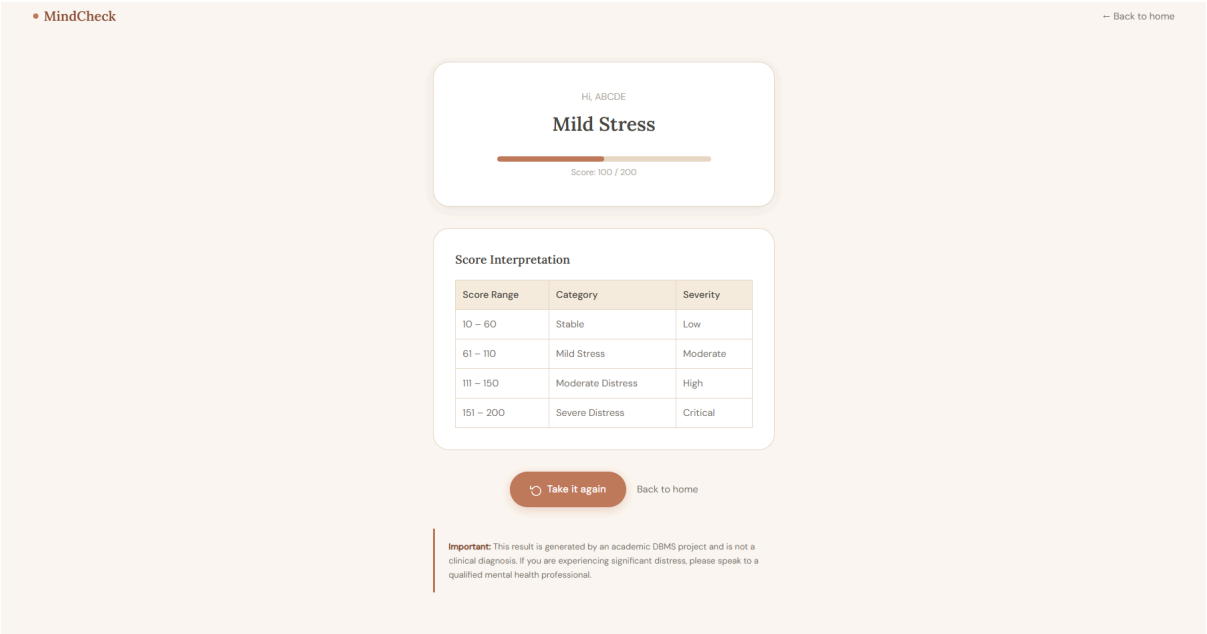

MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 8 of 10
70% complete
QUESTION 8 OF 10
Do you feel hopeless about the future?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →


MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 9 of 10
80% complete
QUESTION 9 OF 10
Do you experience mood swings?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Next question →


MindCheck
← Back to home
Mental Health Assessment
Answer each question honestly based on how you have been feeling over the past two weeks.
Question 10 of 10
90% complete
QUESTION 10 OF 10
Do you avoid social interaction?
Never
Rarely
Sometimes
Often
Almost always
← Previous
Review answers →



7.1.3 Result Page



7.2 Screenshots of Database

7.2.1 Mental Health Reference

ref_id	score_range_min	score_range_max	status_category
1	10	60	Stable
2	61	110	Mild Stress
3	111	150	Moderate Distress
4	151	200	Severe Distress
NULL	NULL	NULL	NULL

7.2.2 Questions

question_id	question_text	max_score
1	Do you feel stressed frequently?	20
2	Do you have trouble sleeping?	20
3	Do you feel anxious without reason?	20
4	Do you feel emotionally drained?	20
5	Do you lose interest in activities?	20
6	Do you feel irritated often?	20
7	Do you struggle to concentrate?	20
8	Do you feel hopeless?	20
9	Do you experience mood swings?	20
10	Do you avoid social interaction?	20

7.2.3 Users

user_id	user_name	user_email	created_at
1	Rahul Sharma	rahul@example.com	2026-03-21 08:29:23
2	Ananya Iyer	ananya@example.com	2026-03-21 08:29:23
4	Test User	testuser123@gmail.com	2026-04-06 23:17:30
5	iy	kh	2026-04-08 14:17:33
6	sdf	sdf	2026-04-08 14:39:44
7	sf	fs	2026-04-08 14:51:38
9	Banana Fish	bananaphish@gmail.com	2026-04-10 14:07:55
10	Test User	test@example.com	2026-04-16 15:36:54
11	New User	new@example.com	2026-04-16 15:37:25
14	ABCDE	ABCDE@123.com	2026-04-26 10:31:42

7.2.4 Responses

response_id	assessment_id	question_id	score
1	1	1	10
2	1	2	12
3	1	3	11
4	1	4	13
5	1	5	9
6	1	6	10
7	1	7	11
8	1	8	12
9	1	9	10
10	1	10	11

7.2.5 Assessment

assessment_id	user_id	submission_date	total_score
1	1	2026-03-21 08:29:24	136
2	2	2026-03-21 08:29:24	NULL
3	4	2026-04-06 23:17:30	100
4	5	2026-04-08 14:17:33	2
5	6	2026-04-08 14:39:44	6
6	7	2026-04-08 14:51:38	18
7	9	2026-04-10 14:07:55	18
8	10	2026-04-16 15:36:54	NULL
9	14	2026-04-26 10:31:42	18

References

1. W3Schools SQL Tutorial.
2. GeeksforGeeks – Database Management System (DBMS).